

Hybrid-Reluctance Actuated Fast Steering Mirror

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Experiments

1.1 PSD Signal Processing

In the first step, the position signal from the PSD (Photosensitive Device) is implemented in SIMULINK. The device outputs four photocurrents from its two-dimensional sensor. The laser spot coordinates are calculated using the formula provided in the lab script:

$$x = \frac{L}{2} \cdot \frac{(i_{x2} + i_{y1}) - (i_{x1} + i_{y2})}{i_{x1} + i_{x2} + i_{y1} + i_{y2}}, \quad y = \frac{L}{2} \cdot \frac{(i_{x2} + i_{y2}) - (i_{x1} + i_{y1})}{i_{x1} + i_{x2} + i_{y1} + i_{y2}}, \quad (1.1)$$

with $L = 9 \text{ mm}$ being the sensor length.

We implemented these equations as a MATLAB SIMULINK block. We verified the correct workings of our implementation via the oscilloscope. To utilize the available signal range from -1.65 V to $+1.65 \text{ V}$, we performed a gain adjustment in SIMULINK. The gain value was determined empirically to maximize the dynamic range of the PSD readout.

1.2 System Identification

To design a controller for the FSM, we first need to identify the system. Under the assumption that both axes are decoupled, we can identify them as two separate systems. For this, we measured two frequency responses, one for each axis. They are given in Figure 1.1 and Figure 1.2. Examining both Bode plots, we can see a resonance at about 100 Hz , both for the x- and y-axis frequency response. Unfortunately, this will limit our control bandwidth, as we have to design our controller in a way to stay above this resonance.

1.3 Controller Design

Based on the measured system dynamics, separate PID controllers were designed for the x- and y-axis of the FSM. Since the axes are symmetrical and show similar Bode plots, the same controller structure was used for both directions. We chose a crossover frequency above the identified resonances

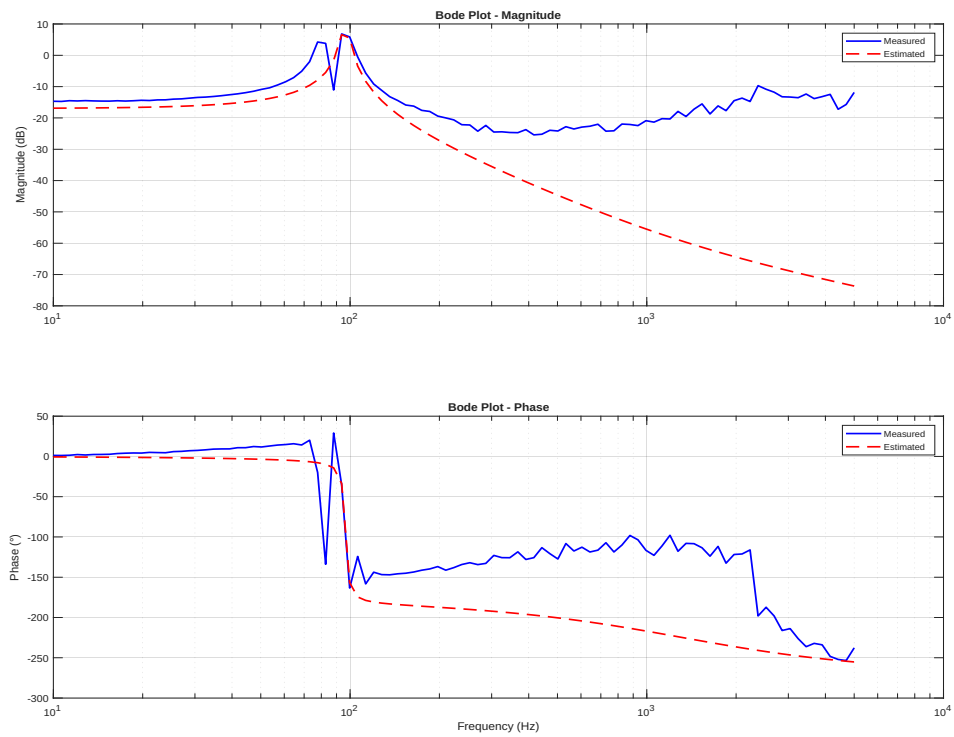


Figure 1.1: Measured Bode plot of the x-axis (input: current reference, output: x-position)

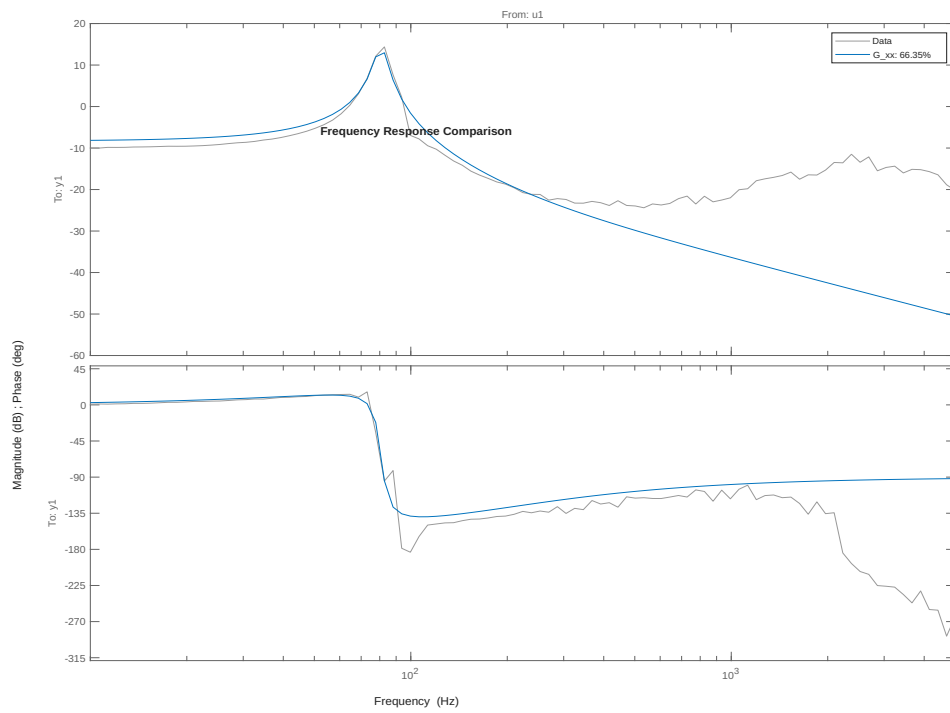


Figure 1.2: Measured Bode plot of the y-axis (input: current reference, output: y-position)

from Section 1.2. This gives our controller a bandwidth starting from just above the resonance frequency up to its crossover frequency.

1.3.1 Alpha-Tuning Method

To tune the PID gains, we chose the alpha-tuning method. Here, the whole controller dynamics are governed by one tuning parameter α . Concretely, the gains for the PID controller

$$C(s) = K_p + \frac{K_i}{s} + K_d s, \quad (1.2)$$

are chosen like this:

$$K_p = \frac{1}{\alpha \cdot |G(j\omega_c)|}, \quad (1.3)$$

$$K_i = \frac{\omega_c}{\alpha^2} \cdot K_p, \quad (1.4)$$

$$K_d = \frac{\alpha}{\omega_c} \cdot K_p. \quad (1.5)$$

We chose a crossover frequency of

$$f_c = 160 \text{ Hz}, \quad \Rightarrow \omega_c = 1005 \frac{\text{rad}}{\text{s}}, \quad (1.6)$$

and set $\alpha = 3$. After discretization with the given sample time of $T_s = 40 \mu\text{s}$, we arrived at

$$K_p^d = \frac{T_s}{2} \cdot K_p = \frac{40 \mu\text{s}}{2} \cdot 1.668524 = 3.337048 \times 10^{-8}, \quad (1.7)$$

$$K_i^d = \frac{T_s}{2} \cdot K_i = \frac{40 \mu\text{s}}{2} \cdot 186.375897 = 3.72751794 \times 10^{-8}, \quad (1.8)$$

$$K_d^d = \frac{T_s}{2} \cdot K_d = \frac{40 \mu\text{s}}{2} \cdot 0.004979 = 9.958 \times 10^{-11}. \quad (1.9)$$

We discovered a bug in our code, where f_c instead of ω_c was used. We therefore recalculated the gains after the lab.

$$K_p^d = \frac{T_s}{2} \cdot K_p = \frac{40 \mu\text{s}}{2} \cdot 2.123296 = 4.246592 \times 10^{-8}, \quad (1.10)$$

$$K_i^d = \frac{T_s}{2} \cdot K_i = \frac{40 \mu\text{s}}{2} \cdot 237.174486 = 4.74348972 \times 10^{-6}, \quad (1.11)$$

$$K_d^d = \frac{T_s}{2} \cdot K_d = \frac{40 \mu\text{s}}{2} \cdot 0.006336 = 1.2672 \times 10^{-10}. \quad (1.12)$$

As this error was only discovered after the laboratory, we did all future tasks with these incorrect calculations. However, since the mix-up from f_c to ω_c only contributes a factor in the calculations, the error was compensated by our chosen crossover frequency and α . Also, only the gains for the x-axis

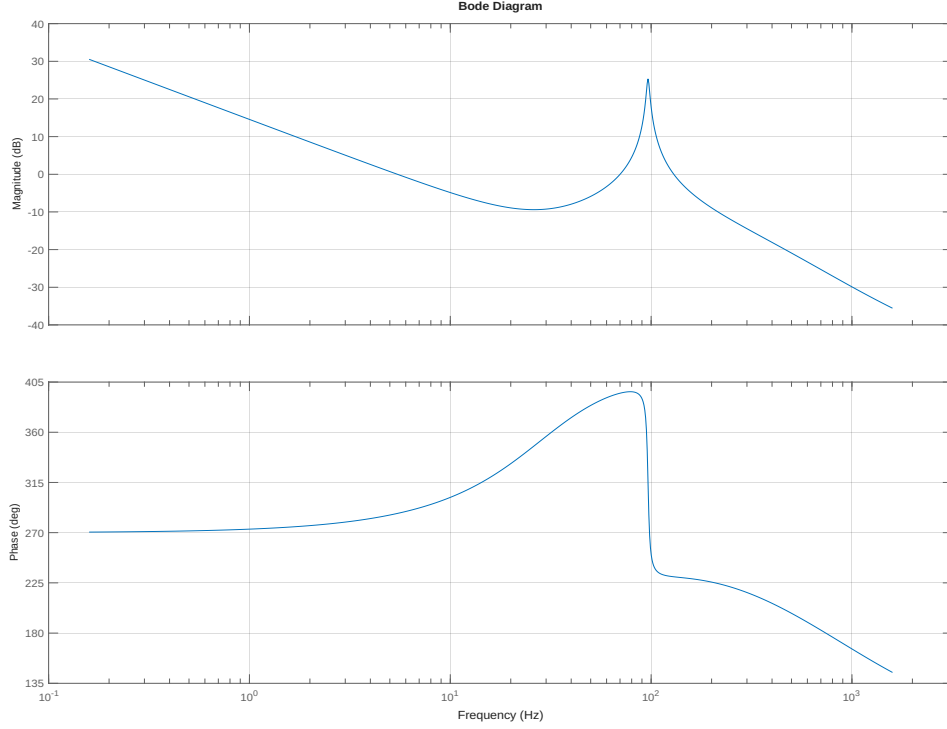


Figure 1.3: Open loop Bode plot for the x-axis controller with plant.

are given here. For the y-axis method and values are similar.

In Figure 1.3 and Figure 1.5, the Bode plot of the open loop are depicted. Additionally, in Figure 1.4 and Figure 1.6, the estimated step responses are given. The two step responses verify the stability of our control loops and also reveal a relatively conservatively tuned PID controller for both axes. In the open loop Bode plots, we see a reduced bandwidth of our desired $f_c = 160$. This is expected in the alpha-tuning method, as for $\alpha \geq 1$, the magnitude correction with K_p gets damped by α .

The controllers were then implemented in SIMULINK and uploaded to the STM32 board.

1.4 Evaluation of Control Performance

1.4.1 System Sensitivity

First, we want to determine the Sensitivity function of our system. For this, we first measure two frequency responses. One with the controller turned off

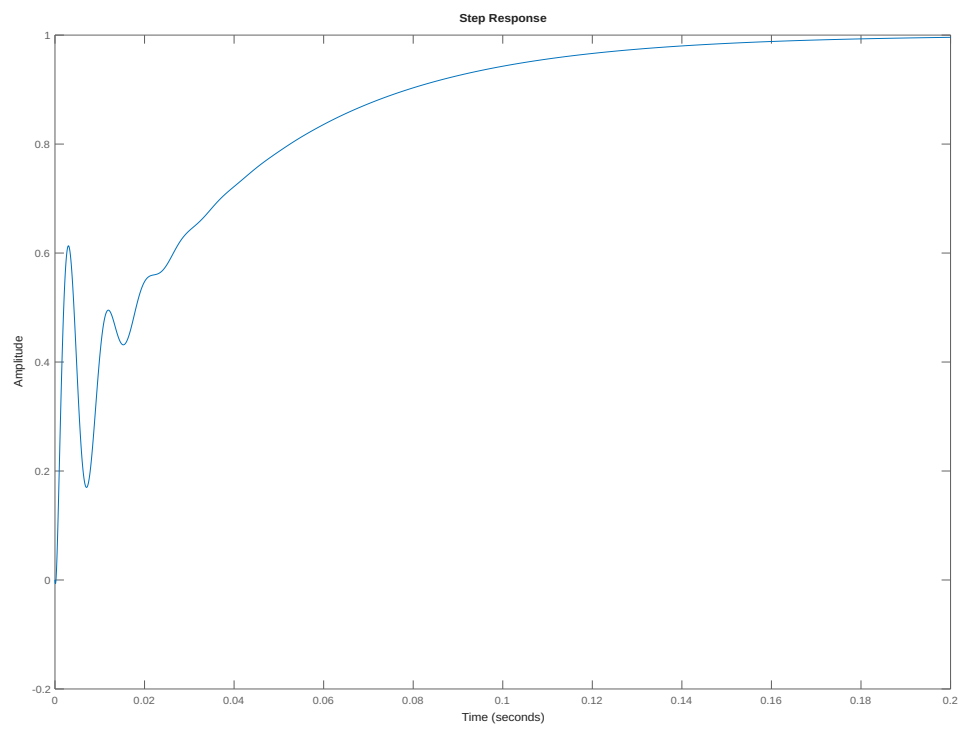


Figure 1.4: Simulated step response of closed loop controller with plant for the x-axis.

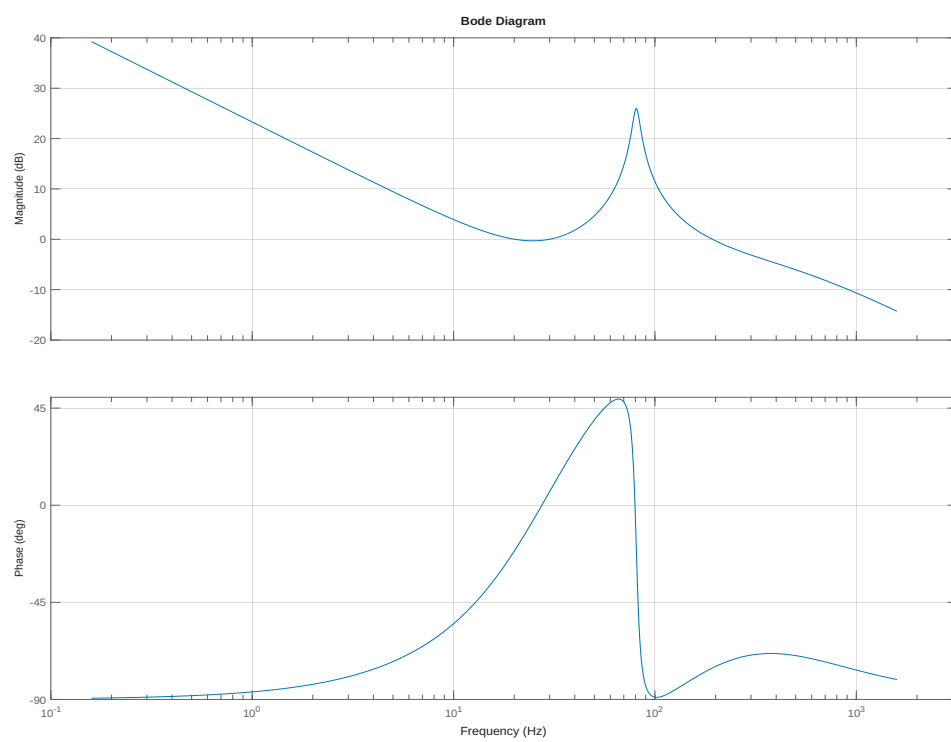


Figure 1.5: Open loop Bode plot for the y-axis controller with plant.

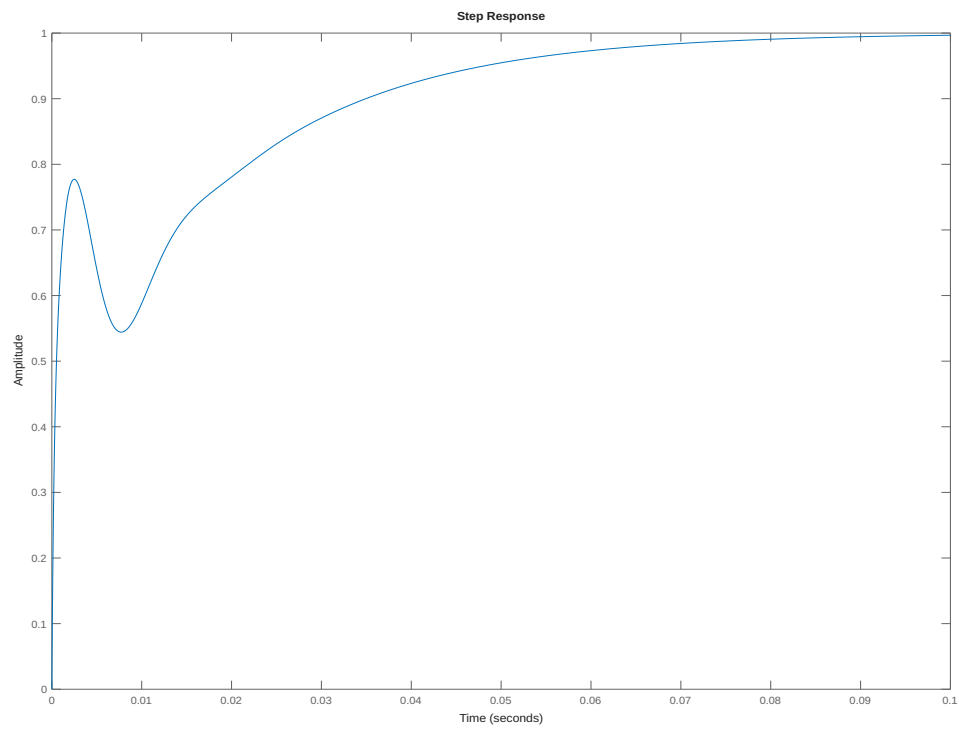


Figure 1.6: Simulated step response of closed loop controller with plant for the y-axis.

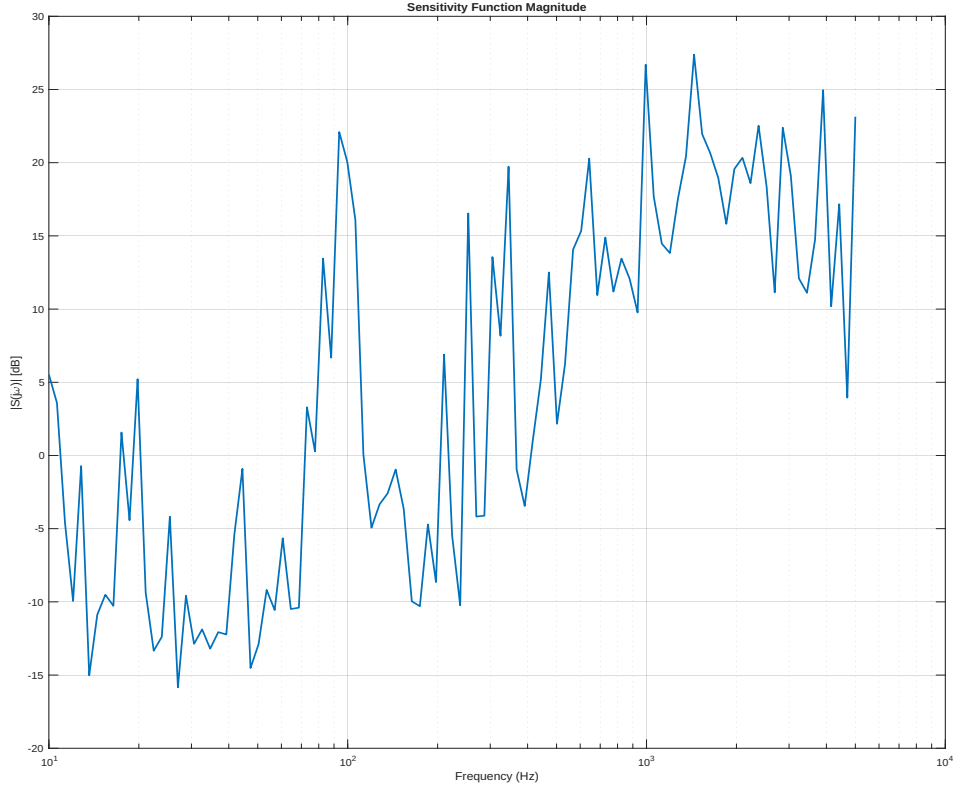


Figure 1.7: Calculated sensitivity from two Frequency sweeps for the y-axis.

and one with the controller turned on. Then we can calculate the Sensitivity

$$S(j\omega) = \frac{T_{\text{open}}(j\omega)}{T_{\text{closed}}(j\omega)}. \quad (1.13)$$

The resulting Sensitivity is plotted in Figure 1.7 for the y-axis. Examining the Sensitivity, the resonance around 100 Hz is clearly visible again. After the resonance is over, the Sensitivity goes down to about -10 dB, indicating a decent working controller inside the control bandwidth. However, ideally we would expect the Sensitivity to be even lower in that frequency range.

1.4.2 Disturbance Rejection

To measure the disturbance rejection, we applied a low frequency sine wave at the voice coil actuator. This moves the sensor up and down. The controller then ideally follows the movement of the sensor. The results can be seen in Figure 1.8. From this we conclude decent but not perfect controller performance.

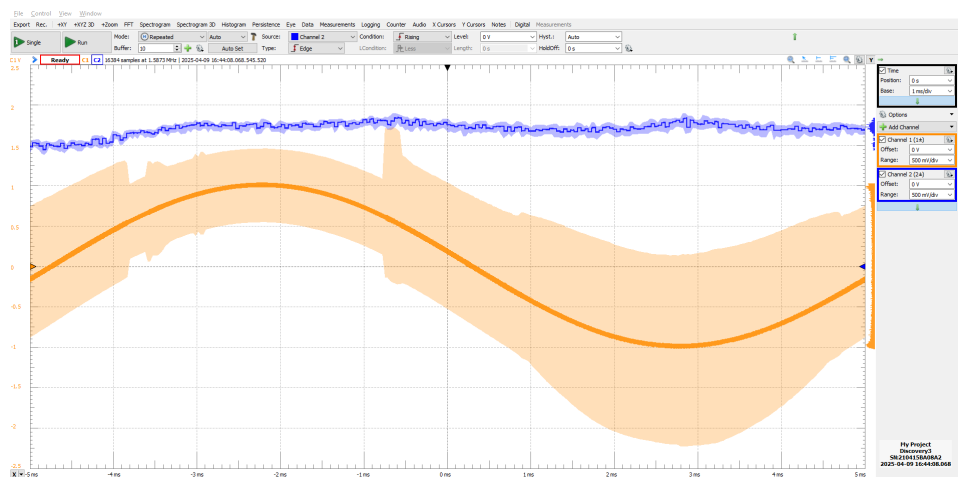


Figure 1.8: In yellow the applied sine wave is plotted and in blue the resulting error.

Report Questions

2.1 System Bode Plots

The Bode plots are given in Figure 1.1 and 1.2. As already discussed, both directions exhibit a clear resonance at about 100 Hz. This can be attributed to internal vibration modes of the FSM. This resonance of course negatively affects our controlled system behavior because we effectively cannot operate the FSM around this resonance. This was taken into account in the controller design by placing the crossover frequency significantly over the resonance peak, giving us a control bandwidth between the resonance and the crossover frequency.

2.2 Sensitivity Discussion

The Sensitivity function is plotted in Figure 1.7. After the resonance, the Sensitivity is reasonably low up until 200 Hz. This indicates reasonable but not great disturbance rejection up until that point. This is consistent with the alpha tuning method and our chosen crossover frequency of $f_c = 160$ Hz.

Conclusion

In this lab exercise, the dynamics of a hybrid-reluctance actuated Fast Steering Mirror (FSM) system were identified and controlled.

The position signal from the PSD was successfully implemented in SIMULINK and calibrated to make full use of the available DAC output range. A frequency-domain system identification was conducted using a sine sweep from 1 Hz to 5 kHz, revealing resonance behavior around 100 Hz for both axes.

Based on the identified plant, a tamed PID controller was designed and tuned using frequency-domain methods to achieve sufficient phase and gain margins while suppressing disturbances effectively. The controller was implemented on the STM32 microcontroller and verified experimentally.

Overall, we were able to implement stable control loops for both the x- and y-axis with decent disturbance rejection performance. Further tuning of the controller parameters would likely yield even better performance. A more sophisticated tuning method might also yield better results.